

# Real-Time Stress Mood Detection Using Deep Learning And Webcam-Based Facial Analysis

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**Abstract:** People are often subjected to increased psychological stress and emotional instability in today's academic and professional settings, which has led to a significant need for easily accessible mental health monitoring services. In order to identify emotional states and gauge stress levels, this work presents an automated framework that analyzes live video captured by webcams. In order to differentiate between seven different affective categories—anger, disgust, fear, happiness, sadness, surprise, and neutral expressions—the implementation makes use of convolutional neural networks trained on greyscale facial imagery (48×48 resolution). Random oversampling techniques were used to correct dataset imbalances, and Haar cascade algorithms were used for real-time processing in facial localization. Comprehensive visualization elements such as probabilistic emotion distributions, dynamic stress gauges, and aggregate stress metrics calculated over monitoring sessions are fed by detected emotions. One unique feature is an emotionally responsive conversational agent that provides individualized wellness adviced by adjusting interaction strategies based on detected affective states. Performance evaluation shows effective real-time operation and satisfactory classification accuracy, indicating viability for realistic affective computing deployments in mental wellness applications.

**Keywords**—Facial expression recognition, stress detection, real-time systems, deep learning, human–computer interaction, chatbot.

## I. INTRODUCTION

Human functioning is being negatively impacted by psychological stress in a number of ways, including productivity at work, academic success, emotional control, and general quality of life. Growing demands on education and career expectations are imposed by modern society, which frequently results in emotional disorders that worsen undetected until they become clinical conditions [11,12]. Traditional assessment techniques, such as self-administered questionnaires, psychological inventories, and clinical interviews, have inherent flaws like episodic administration, subjective bias, and a lack of continuous monitoring capabilities [10,11]. These drawbacks show how urgently automated surveillance systems that can surreptitiously and continuously track emotional states in naturalistic environments are needed.

One of the most common and easily understood forms of nonverbal communication is facial expression. Affective signals that correspond to underlying emotional experiences like frustration, contentment, anxiety, or astonishment are naturally conveyed by human faces [12]. The development of useful systems for emotion recognition without the need for specialized hardware has been made possible by the widespread use of consumer-grade webcams as well as significant advancements in deep learning techniques and computer vision algorithms. Convolutional neural networks consistently achieve state-of-the-art performance across common emotion recognition benchmarks, making them particularly effective architectures for learning discriminative facial features [1,2,3,7].

An integrated platform for real-time affective state monitoring via webcam-based facial analysis driven by deep learning, building on these technological underpinnings [5,6,9,11]. A convolutional neural network that has been trained to distinguish between seven basic emotional categories anger, disgust, fear, happiness, sadness, surprise, and neutral is used in the architecture. In order to address skewed distributions found in training data, preprocessing techniques use random oversampling to guarantee balanced representation across emotion classes. Facial localization within video frames is accomplished using Haar cascade classifiers, allowing efficient real-time detection [8].

To help users better understand their affective patterns, the system incorporates a number of interactive visualization features that go beyond basic emotion classification. These include instantaneous stress indicators through dynamic gauge displays, session-aggregate emotion probability distributions shown by bar graphs, and computed overall stress percentage [1,7]. An especially creative addition is the incorporation of an affect-aware conversational agent that adapts response strategies according to recognized emotional states. This chatbot offers contextually relevant content, such as calming techniques, encouraging messages, and supportive dialogue, that is customized to the user's current mood.

The proposed framework combines CNN-based emotion

recognition, real-time facial detection, stress quantification algorithms, and affect-responsive conversational support to provide comprehensive emotional wellness monitoring using only standard webcam hardware. This approach is fully autonomous, non-intrusive, and capable of continuous deployment in everyday environments, in contrast to conventional approaches that require invasive physiological sensors or manual self-report procedures [13,14].

One of the primary contributions of this work is a deep learning architecture for facial emotion recognition trained on balanced datasets obtained through random oversampling techniques. • The development of a real-time processing pipeline that combines CNN-based emotion classification from webcam streams with Haar cascade facial detection. • The development of a stress estimation system that consists of a stressometer, an emotion probability distribution graph, and a session-level stress percentage. • Building a multi-component stress visualization system that includes session-level stress aggregation, emotion probability distributions, and dynamic stress gauges. • The incorporation of a chatbot with emotional intelligence that can modify conversational responses in response to real-time emotion detection. • The implementation of a completely autonomous real-time monitoring system appropriate for applications involving mental wellness and user-centric affective computing [4,8].

The rest of the paper is organized as follows: Section II reviews relevant literature on stress detection and emotion recognition; Section III describes the system architecture; Section IV describes the dataset, preprocessing techniques, and class balancing strategies; Section V presents the CNN model architecture and training methodology; Section VI expounds on the stress estimation algorithms and real-time inference pipeline; Section VII reports experimental results; and Section VIII concludes with future research directions.

## II. RELATED WORK

Affective computing research spans several interrelated fields, such as emotion-responsive conversational systems, stress measurement, and facial expression analysis. Relevant contributions from these fields are reviewed in this section.

### A. Facial Expression Recognition

Using the Facial Action Coding System (FACS), Ekman and Friesen's seminal work created methodical frameworks for examining facial movements [1]. Local Binary Patterns (LBP), Gabor wavelets, and Haar-like features were examples of manually created feature descriptors used in early computational methods for facial expression recognition [2,3]. Although these techniques worked well in controlled settings, they were not very resilient to changes in head orientation and lighting. Deep learning significantly improved recognition performance. While Goodfellow et al. introduced maxout networks for robust feature learning [5], Liu et al. showed CNN's efficacy for emotion classification tasks [4]. Inception-based deep architectures were later proposed by Mollahosseini et al., who achieved better outcomes on common FER benchmarks

[6]. These developments make CNNs an excellent choice for applications involving real-time facial expression recognition.

### B. Stress Detection and Affective Monitoring

Physiological measurements such as heart rate variability (HRV), electro dermal activity (EDA), electroencephalography (EEG), and respiration patterns are the mainstay of traditional stress detection techniques [7-9]. These methods require wearable sensing devices, which limit their practical deployment for continuous daily monitoring even though they achieve reliable accuracy. A non-contact alternative is provided by facial analysis. Sriramprasad et al. used motion patterns and facial micro expressions to predict stress [10]. The potential of facial features was highlighted by Giannakakis et al.'s thorough survey of behavioral and visual cues for stress inference [11]. Nevertheless, a lot of current systems lack useful deployment frameworks or real-time operational capabilities.

### C. Affective Conversational Agents

Research on affective computing includes conversational systems that adjust responses based on the emotions of the user. Kismet and other early researchers looked into expressive emotional interactions [12]. In order to produce sympathetic responses, modern neural conversational models use sentiment embedding's and emotional context [13,14]. Despite these developments, there is still a large gap in affect-aware human-computer interaction since few systems directly incorporate real-time facial expression recognition into chatbot architectures.

### D. Research Gap

Although significant advancements have been made in affect-aware chatbots, stress detection, and facial expression recognition, current systems usually: rely on physiological sensors or laboratory conditions; are unable to infer emotions in real time; do not include visual stress indicators or thorough emotional summaries; and infrequently combine facial expression recognition with adaptive conversational agents. By offering a unified real-time framework that combines webcam-based facial expression recognition, stress quantification, visualisation dashboards, and a mood-responsive chatbot, the suggested system overcomes these drawbacks.

## III. SYSTEM ARCHITECTURE

The suggested Real-Time Stress and Mood Detection System is a modular end-to-end pipeline that analyses live webcam video, uses convolutional neural networks to classify emotions, measures stress levels, and uses mood-adaptive chatbots and visualization interfaces to engage users. The four main layers of the architecture are application/visualization, deep learning inference, preprocessing, and input acquisition.

### A. Input Acquisition Layer

This part uses the OpenCV library to record live video streams from the user's webcam. Every frame is continuously collected and sent in real time for further processing. By using standard webcam hardware, platform independence is

ensured and deployment on common consumer devices is made possible without the need for specialized equipment.

### B. Face Detection and Preprocessing Layer

The system locates facial regions within each video frame using Haar cascade classifiers. When a face is identified, the area of interest is cropped, converted to greyscale, resized to 48 by 48 pixels, and normalized to the interval [0,1]. These preprocessing steps improve prediction robustness under different lighting conditions and guarantee compatibility with CNN input specifications.

### C. Deep Learning Inference Layer

A trained convolutional neural network that can identify seven emotions—anger, disgust, fear, happiness, sadness, surprise, and neutral—is fed preprocessed facial images. A softmax layer at the end of the CNN architecture creates a seven-dimensional probability distribution. The predicted class label is chosen to be the emotion with the highest probability. TensorFlow/Keras is used in model deployment to load pre-trained weights and architecture configurations for effective real-time performance.

### D. Stress Estimation and Mapping Layer

Stress estimation is derived from a mapping function that interprets predicted emotion probabilities, whereas the CNN predicts discrete emotion labels. While happiness and neutral states are associated with lower levels of stress, emotions like anger, fear, and sadness are associated with higher levels of stress. Instantaneous stress scores are computed per frame, whereas session-level stress is calculated using aggregated emotion probabilities over the monitoring period.

### E. Visualization and User Interaction Layer

This layer offers unambiguous real-time feedback through a number of interactive components: • **Stress-O-Meter**: A dynamic gauge that displays the level of stress. • **Emotion Probability Bar Graph**: This graph displays how the emotions that were identified during the session were distributed. • **Overall Stress Percentage**: Indicates how much stress has built up over time. • **Suggestion Panel**: It provides brief wellness suggestions based on the user's current emotional state. • **Mood-Aware Chatbot**: Generates conversational responses that are modified according to detected emotions and stress levels. These elements come together to create a comprehensive real-time emotional monitoring system that offers users interactive assistance and useful insights.

## IV. DATASET AND PREPROCESSING

The performance of facial emotion recognition systems depends on the caliber, variety, and balance of training data.

This section covers our system's train-test partitioning, class balancing method, preprocessing pipeline, and dataset structure.

### A. Dataset Structure and Label Mapping

The dataset has a structure based on directories with Each type of emotion is stored in a separate folder. By scanning directory names, seven classes of emotions—including anger, disgust, fear, happiness, sadness, surprise, and neutral—are automatically recognized. OpenCV is used to read images, and NumPy is used to store them in arrays. A dictionary that links folder names to integer indices is used to programmatically assign labels. These automated techniques eliminate human labeling errors and ensure consistency. Fig. 1 displays sample counts for each of the seven emotions. classes prior to preprocessing, revealing stark differences between categories

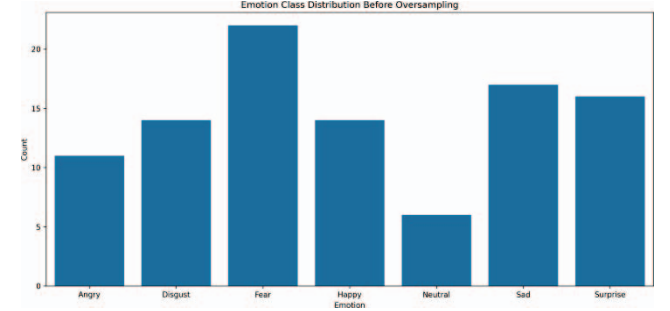


Fig. 1. Distribution of emotion classes before applying oversampling. Significant imbalance is observed among the seven categories.

### B. Image Preprocessing

To guarantee consistent CNN input, each image goes through a number of preprocessing steps: • **Greyscale Conversion**: To cut down on computational complexity, all images are converted to greyscale. • **Resizing**: To match the dimensions of the CNN input, each image is resized to 48 by 48 pixels. • **Flattening**: Before oversampling, images are reshaped into 2304-dimensional vectors to satisfy RandomOverSampler requirements. The range of facial features captured in the dataset is illustrated by visual samples of preprocessed images that represent each emotion class.

### C. Class Balancing Through Random Oversampling

A preliminary analysis showed an imbalance in the dataset, with some emotion categories being noticeably under-represented. Overfitting to majority classes is a risk when deep learning models are trained on unbalanced data. Random Oversampling from the imbalanced-learn library was used to solve this. This method creates a balanced dataset by randomly duplicating samples from minority classes until all classes have an equal number of examples:

$$X_{\text{resampled}} \in \mathbb{R}^{N' \times 2304}, \quad y_{\text{resampled}} \in \mathbb{Z}^{N'} \dots \dots (1)$$

Fig. 2 shows the class distribution after oversampling where all emotional categories now contain an equal number of samples.

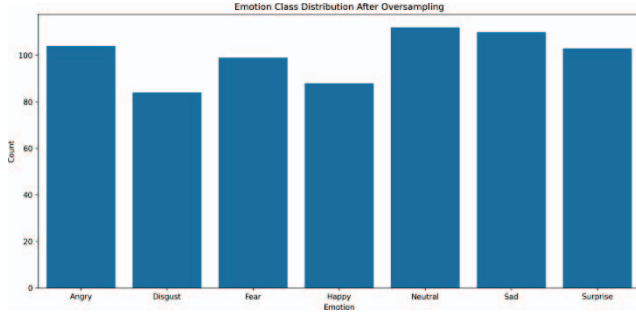


Fig. 2. Distribution of emotion classes after applying Random Oversampling. All seven classes are balanced.

#### D. Normalization and Reshaping

After oversampling, pixel values are converted to 32-bit floating-point format and then divided by 255 to normalise them to the range [0,1]. After that, flattened vectors are reshaped into 4D tensors that can be fed into CNNs:

$$X_{\text{final}} \in \mathbb{R}^{N \times 48 \times 48 \times 1} \dots (2)$$

Normalization accelerates convergence and improves training stability.

#### E. Train-Test Split and One-Hot Encoding

To guarantee reproducibility, the balanced dataset is divided into training and testing subsets using a 90/10 ratio and a fixed random seed. To enable the softmax output layer, labels are one-hot encoded:

$$y_{\text{train}} \in \{0, 1\}^{N_{\text{train}} \times 7}, \quad y_{\text{test}} \in \{0, 1\}^{N_{\text{test}} \times 7} \dots (3)$$

This format is standard for multi-class classification in deep learning frameworks.

### V. MODEL ARCHITECTURE AND TRAINING

A deep convolutional neural network optimized for real-time inference on 48x48 grayscale facial images is used in the suggested facial emotion recognition system. By balancing expressive feature learning with computational efficiency, the architecture guarantees accurate predictions while maintaining the processing speeds needed for webcam-based applications.

#### A. CNN Usage Overview

The CNN is composed of five convolutional blocks and fully connected layers. Each convolutional layer is paired with batch normalization and ReLU activation to promote faster convergence and stable gradient flow. While MaxPooling layers progressively reduce spatial dimensionality, Dropout layers prevent feature co-adaptation, which reduces overfitting.

#### C. Model Compilation

The model is compiled using the Adam optimizer with learning rate 0.0001. Categorical cross entropy serves as the

#### B. Layer-by-Layer Description

The CNN consists of the following components:

accepts 48x48 grayscale images that have been normalized.

- **Input Layer:**
- **Convolution Block 1:** Conv2D with 32 filters (3x3), dropout (0.25), batch normalization, and ReLU activation.
- **Convolution Block 2:** Conv2D with 64 filters (3x3), MaxPooling (2x2), Batch Normalization, and ReLU activation.
- **Convolution Block 3:** ReLU activation, batch normalization, and Conv2D with 64 filters (3x3).
- **Convolution Block 4:** MaxPooling (2x2), ReLU activation, batch normalization, and Conv2D with 128 filters (3x3).
- **Convolution Block 5:** MaxPooling (2x2), ReLU activation, batch normalization, and Conv2D with 128 filters (3x3).
- **Flatten Layer:** creates dense vectors from extracted feature maps.
- **Dense Layer:** 250 neurons in a fully connected layer with ReLU activation and Dropout (0.5).
- **Output Layer:** Dense(7) for seven-class emotion classi-

$$L = - \sum_{i=1}^7 y_i \log(\hat{y}_i) \dots (4)$$

Primary performance metric is classification accuracy.

#### D. Training Procedure

The balanced and normalized dataset outlined in Section IV is used to train the CNN for 35 epochs. Batch normalization stabilizes training, while dropout prevents overfitting. Accuracy and loss are monitored during training and validation to ensure proper convergence.

Fig. 3 displays the accuracy curves for training and validation, while Fig. 4 displays the corresponding loss curves. The effectiveness of the model architecture is demonstrated by these curves, which exhibit steady convergence with minimal overfitting.

#### E. Model Saving and Deployment

To enable smooth loading in the real-time inference pipeline, the final trained CNN model is saved in both HDF5 and JSON+weights formats. Future extensions, such as retraining, fine-tuning, and deployment in interactive applications like the integrated mood-aware chatbot, are made possible by this modular saving approach.

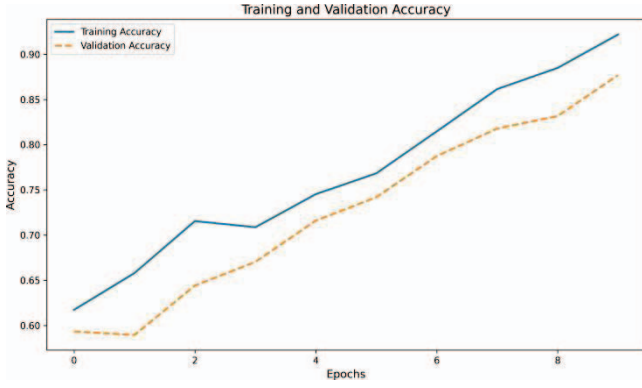


Fig. 3. accuracy over 35 epochs of training and validation. Without any indications of overfitting, the model exhibits robust and steady convergence.

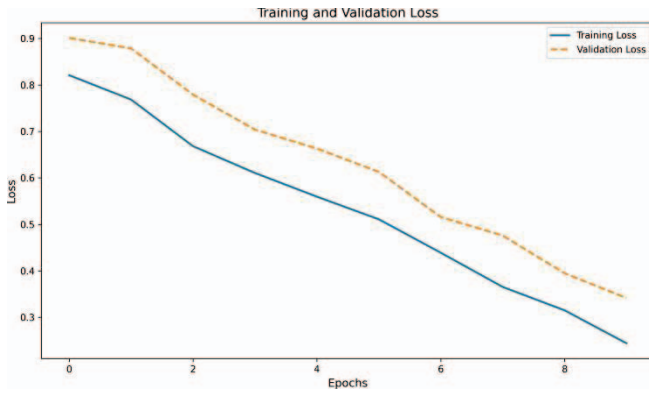


Fig. 4. 35 epochs of training and validation loss. Consistent optimization during the training process is confirmed by the declining loss curves.

## VI. REAL-TIME INFERENCE AND STRESS ESTIMATION

Through live webcam feeds, the real-time component allows for ongoing monitoring of user emotional states. Video acquisition, face detection, emotion classification, stress score computation, and user interaction through mood-aware chatbots and visualization dashboards are all included in the processing pipeline.

### A. Real-Time Video Capture

The OpenCV library is used to record live video frames from the webcam on the device. Users can receive immediate emotional feedback because each frame is streamed to the processing pipeline with little latency.

### B. Face Detection Using Haar Cascade

Haar cascade classifiers trained on frontal face samples are used to identify facial regions. The classifier uses feature matching to identify facial regions for every grayscale frame. The detected region of interest is resized to 48×48 pixels to match CNN input dimensions.

### C. Emotion Prediction

After being extracted, the face region is reshaped into a 4D tensor, normalized to [0,1], and fed through the trained CNN. A seven-dimensional probability vector that represents the seven emotion classes is produced by the model. The predicted label for that frame is chosen to be the emotion with the highest probability.

### D. Stress Score Computation

Stress estimation uses a mapping function based on emotion probabilities. Emotions like happiness, neutrality, and surprise have a negative effect on stress, whereas emotions like anger, fear, and sadness have a positive effect. The following formula is used to determine the instantaneous stress score:

$$S_t = P(\text{anger}) + P(\text{fear}) + P(\text{sadness}), \dots \dots (5)$$

where  $S_t$  denotes the stress level at time  $t$ . The session-level stress score is computed as the average across all frames:

$$S_{\text{session}} = \frac{1}{T} \sum_{t=1}^T S_t \dots (6)$$

### E. Visualization Dashboard

The user interface includes:

- **Stress-O-Meter:** shows the stress score in real time.
- **.xtbfEmotion Probability Graph:** visualizes the likelihood of each emotion throughout the session.
- **Overall Stress Percentage:** summarizes stress at the session level.
- **Wellness Suggestions:** gives advice in real time based on one's emotional state

### F. Mood-Aware Chatbot

To improve user experience and encourage emotional well-being, the system incorporates a chatbot that modifies responses based on the emotional states of the user. Positive emotions allow for more laid-back and interesting interactions, while high-stress or negative emotions lead to sympathetic and encouraging conversation.

### G. System Performance

Practical webcam-based emotion recognition is made possible by the real-time pipeline's low latency and high frame rates. For deployment on common consumer hardware, the lightweight CNN architecture and effective preprocessing remove the need for a GPU.

## VII. EXPERIMENTAL RESULTS AND DISCUSSION

The holdout test set was used to assess the trained convolutional neural network's capacity for generalization across seven emotion categories. Accuracy metrics, loss values, per-class performance analysis, and visual inspection using confusion matrices are all included in evaluation.

### A. Quantitative Evaluation

With low loss values, the model achieves competitive accuracy on the test set. The classification report confirms that all categories were successfully separated by providing comprehensive precision, recall, and F1-scores for each emotion class.

### B. Confusion Matrix Analysis

The relationships between true and predicted labels are depicted in the confusion matrix. High true-positive counts for the majority of emotion classes are indicated by strong diagonal elements. Misclassifications mostly happen between visually similar expressions, like fear-surprise and anger-disgust.

### C. Training Behavior Discussion

The training and validation curves shown in Fig. 3 and Fig. 4 (Section V) show minimal divergence between training and validation performance and smooth convergence, suggesting appropriate model regularization without obvious overfitting.

### D. Overall Discussion

The suggested CNN architecture effectively distinguishes all seven emotion classes, according to the results. Random Oversampling eliminates dataset bias and produces balanced per-class performance. The model is perfect for real-time webcam-based applications because of its strong accuracy, quick inference speed, and resilience to real-world variations.

## VIII. CONCLUSION AND FUTURE WORK

The suggested real-time stress and mood detection system successfully combines webcam-enabled inference pipelines for ongoing affect monitoring with deep learning-based facial emotion recognition. Reliable preprocessing, class-balancing strategies, and a well-designed convolutional neural network enable strong performance in all seven emotion categories. Probability visualization, stress estimation, and a mood-aware chatbot are all integrated to enhance system usability and provide personalized and interactive wellness experiences.

The system is suitable for daily use because it operates reliably and efficiently in real-time on common hardware. Its stress-o-meter, emotion probability graphs, and session-level summaries offer clear and practical insights into user emotional states.

Although the system functions effectively, there are still certain aspects that could be enhanced. Future studies could focus on expanding the dataset to include more demographic variations, incorporating advanced natural language models to improve chatbot flexibility, and incorporating multimodal emotional cues such as speech prosody or physiological signals. Deployment on mobile or edge devices may also enable accessibility and broader adoption.

All things considered, the proposed system provides a strong foundation for real-time affective computing and facilitates the development of emotionally intelligent tools for human-computer interaction.

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